

Hindi TTS Experiment: Phoneme Clustering Impact on Speech Synthesis

Research Question

Does reducing the G2P output vocabulary from 57 phonemes to 39 phoneme clusters degrade synthesized speech quality when using VITS (Variational Inference with adversarial learning for end-to-end Text-to-Speech)?

Dataset

- **Source:** Hugging Face `SPRINGLab/IndicTTS-Hindi`
- **Licence:** IndicTTS licence (accepted prior to download)
- **Voice selection:** Female recordings only (gender-homogeneous)
- **Target duration:** 1.75 hours (acceptable range: 1.70–1.80 hours)
- **Actual duration:** 1.7494 hours

Sample Statistics

Category	Count
Total candidate rows	1,645
Selected samples	1,031
Excluded (random cap)	614
Training set (90%)	928
Validation set (5%)	53
Test / held-out set (5%)	50

- **Selection seed:** 42
- **Split level:** utterance-level, deterministic
- **No sample appears in more than one split**

Exclusion Criteria

Samples were excluded if they had:

- Empty or missing text
- Duration < 2.5 s or > 20 s
- Corrupt or silent audio
- Incomplete G2P token coverage (any OOV phoneme or unmapped cluster)

Remaining samples exceeding the 1.75-hour target were excluded by random selection (seed 42).

Audio Preprocessing

Parameter	Value
Sample rate	22,050 Hz
Channels	Mono
Silence trimming	Leading/trailing silence removed
Loudness normalization	Applied consistently
Format	WAV (16-bit PCM)

The **same processed WAV files** are used for both baseline and clustered training. No audio difference exists between conditions.

G2P Labeling

Baseline Vocabulary (57 phonemes → 62 tokens)

The baseline uses the project's 57-phoneme inventory from `g2p/phoneme_vocab.json`, plus 5 special tokens (`<pad>` , `<eos>` , `<bos>` , `<blnk>` , `<wb>`), totaling 62 tokens.

Example: `b aa r ax t` → token IDs via whitespace-split tokenizer

Clustered Vocabulary (39 clusters → 44 tokens)

The clustered vocabulary uses 39 phoneme clusters (C0–C38) derived from `g2p/phoneme_cluster_mapping.json`, which maps each of the 57 baseline phonemes to one of 39 clusters. Plus 5 special tokens, totaling 44 tokens.

Example: `C3 C0 C29 C0 C1` → token IDs via whitespace-split tokenizer

Key Constraint

- Cluster tokens are **atomic**: `C10` is one token, not `C + 1 + 0`
- The VITS text-processing path **bypasses any internal phonemizer**
- Word boundaries use the explicit `<wb>` token
- Labels are derived: `text` → `baseline phonemes` → `cluster mapping`
- Both label sets are fully aligned; they differ only in representation

G2P Fallback Statistics

- Primary source: `data/multilingual_g2p_dataset.txt` (exact Hindi word match)
- Fallback: trained baseline G2P model (`models/best_g2p_transformer.weights.h5`)
- Utterances with any unmappable word or phoneme were excluded entirely

Fair-Training Controls

Both VITS models were trained with **identical** settings. The **only** difference is the input vocabulary.

Hyperparameters

Parameter	Value
Architecture	Coqui TTS VITS (single-speaker, from scratch)
Optimizer	AdamW ($\beta_1=0.8$, $\beta_2=0.99$, $\epsilon=1e-9$, <code>weight_decay=0.01</code>)
LR (generator)	0.0002
LR (discriminator)	0.0002
LR scheduler	ExponentialLR ($\gamma=0.999875$)
Batch size	32
Mixed precision	FP16
Max training steps	~15,000 (576 epochs × ~26 steps/epoch)
Checkpoint interval	Every 500 steps
Best checkpoint selection	By best validation loss

Random seed	42 (+ torch seed 54321 for trainer)
CUDNN deterministic	True
CUDNN benchmark	False

Audio Configuration

Parameter	Value
Sample rate	22,050 Hz
FFT size	1,024
Window length	1,024
Hop length	256
Mel channels	80
mel_fmin	0
mel_fmax	None

Hardware & Software

Component	Version
GPU	NVIDIA RTX A5000 (24 GB)
Python	3.10.12
PyTorch	2.4.1+cu124
Coqui TTS	0.27.5
Transformers	4.57.3
OS	Ubuntu 22.04 (WSL2)

Training Results

Training Duration

Model	Training Time	Final Global Step
Baseline (62 tokens)	7.99 hours	15,550
Clustered (44 tokens)	7.93 hours	15,550

Final Evaluation Metrics (Epoch 575)

Metric	Baseline	Clustered	Δ (Clustered – Baseline)
avg_loss_disc	2.6614	2.6691	+0.0077 (+0.3%)
avg_loss_gen	2.0033	2.0140	+0.0107 (+0.5%)
avg_loss_kl	1.4152	1.4122	−0.0030 (−0.2%)
avg_loss_mel	19.7433	19.9201	+0.1768 (+0.9%)
avg_loss_duration	1.7492	1.7476	−0.0016 (−0.1%)
avg_loss_feat	3.3894	3.4479	+0.0585 (+1.7%)

Observation: All losses differ by less than 2%. The clustered model converges to near-identical quality despite 29% fewer input tokens.

Best Checkpoints

Model	Checkpoint Path
Baseline	/mnt/d/tts_models/hi/baseline/vits_hindi_female_baseline-August-31-2026_11+25PM-fb53263/best_model.pth
Clustered	/mnt/d/tts_models/hi/clustered/vits_hindi_female_clustered-September-01-2026_08+04AM-fb53263/best_model.pth

Inference

Held-out Test Set

- 50 samples from the reserved 5% test split
- Each sample synthesized by both models

- Output: `samples/tts_hindi_female/held_out/baseline_XXX.wav` + `clustered_XXX.wav`

Unseen Generalization Set

- 20 new Hindi sentences not present in training, validation, or test data
- Each sentence tokenized via the full G2P pipeline
- All 20/20 sentences were fully tokenizable
- Output: `samples/tts_hindi_female/unseen/baseline_XX.wav` + `clustered_XX.wav`

Evaluation Protocol

Automated MOS (DNSMOS)

Automated MOS scores were computed using Microsoft's DNSMOS (Deep Noise Suppression MOS) predictor via the `speechmos` package. These are **predicted/automated** MOS scores, **not** human MOS scores.

Held-out Set Results

Condition	Mean MOS	Std Dev	95% CI	N
Baseline	3.1117	0.1772	[3.0626, 3.1609]	50
Clustered	3.1557	0.1450	[3.1155, 3.1959]	50

Paired difference (Baseline – Clustered): -0.0439 ± 0.1803

Unseen Set Results

Condition	Mean MOS	Std Dev	95% CI	N
Baseline	3.0129	0.1993	[2.9237, 3.1020]	20
Clustered	3.0184	0.1803	[2.9377, 3.0990]	20

Paired difference (Baseline – Clustered): -0.0055 ± 0.2224

Interpretation

- **Confidence intervals overlap completely** in both evaluation sets
- The clustered model is **statistically indistinguishable** from the baseline

- On the held-out set, the clustered model scores marginally (+0.04) higher
- On the unseen set, the difference is negligible (+0.005)
- These automated scores serve as **supporting evidence** only

Human Evaluation

Human evaluation was not conducted in this phase. Scripts for blinded human evaluation are prepared and ready for deployment:

- `tts/tts_human_eval_prep.py` : generates blinded audio pairs and evaluation forms
- `tts/tts_human_eval_analysis.py` : analyzes collected listener scores with Wilcoxon signed-rank test

Limitations

1. **One language:** Hindi only. Results may not generalize to other languages.
2. **One voice:** Female speaker only. Male voice or multi-speaker scenarios are untested.
3. **Limited data:** 1.75 hours of speech is small for VITS. Results are preliminary.
4. **No human evaluation:** Only automated MOS was computed. Human perceptual evaluation would strengthen claims.
5. **Pilot scope:** This is a proof-of-concept for the phoneme clustering approach, not a production-ready system.
6. **Automated MOS limitations:** DNSMOS was designed for noise suppression quality and may not perfectly capture naturalness/intelligibility differences in synthesized speech.

Important Disclaimers

Older gTTS samples and MOS results in this repository are NOT evidence for this VITS experiment. Those were generated using Google's gTTS as a placeholder and have no bearing on the VITS baseline-vs-clustered comparison.

Automated MOS scores are not a substitute for human listening tests. They are reported as supporting quantitative evidence only.

Reproducibility

All code, configurations, and manifests required to reproduce this experiment are available in the repository:

Component	Path
Data download	tts/tts_data_download.py
Audio preprocessing	tts/tts_audio_preprocess.py
G2P labeling	tts/tts_g2p_labeling.py
Tokenizer	tts/vits_tokenizer.py
Data splitting	tts/tts_split_data.py
VITS training	tts/tts_vits_training.py
Inference	tts/tts_inference.py
Automated evaluation	tts/tts_automated_eval.py
Human eval prep	tts/tts_human_eval_prep.py
Human eval analysis	tts/tts_human_eval_analysis.py
Manifest	data/tts_hindi_female/manifest.csv
Phoneme vocab	g2p/phoneme_vocab.json
Cluster mapping	g2p/phoneme_cluster_mapping.json
Training log	/mnt/d/tts_models/hi/training_log.json
Automated MOS results	results/tts_hindi_female/automated_mos.csv
Automated MOS report	results/tts_hindi_female/automated_mos_report.md

Raw audio, processed audio, model checkpoints, and generated WAV samples are excluded from Git via `.gitignore`.